# Adaptime: Method Overview

Thesis project

Current protocol

**Abstract**

Adaptime studies whether a small fitted adaptor can combine a time-series foundation model with fixed, calendar-aligned retrieval while preserving the official TIME test protocol. This note defines the current univariate `full_ridge_shared` formulation; it reports no empirical result.

## 1 Chronological protocol

Each dataset and forecast term is divided into a fixed datastore, adaptation training, adaptation validation, and the untouched official TIME test interval. The first three regions end strictly before the test. Ridge candidates are fit once on adaptation training, $K$ and $\alpha$ are selected on adaptation validation, and the selected coefficients are frozen for test. There is no train-plus-validation refit.

For a query time $t$, let $d_{\max}$ be the fixed datastore endpoint and $P$ the dataset period. The latest eligible neighbor date is

$$d_t = \min(t - H, d_{\max}) - \big(\min(t - H, d_{\max}) - t\big) \bmod P. \tag{1}$$

Earlier candidates walk backward from $d_t$ by a datastore stride that is a multiple of $P$. This keeps the datastore fixed while aligning its phase to the query.

Let $q_t$ be the finite fraction of a query history. The default gate requires $q_t \geq 0.8$ before retrieval. Datastore histories satisfy the same threshold, and their complete retrieved future must be finite. A query without $K$ valid chronological neighbors is ineligible rather than an experiment-ending error.

## 2 Full shared ridge

For each query, $V$ is the vanilla foundation forecast and $C$ is the forecast obtained when the $K$ retrieved neighbor trajectories are supplied as covariates. For neighbor $i$, $Y_i$ is its observed horizon and $N_i$ is the foundation model's vanilla forecast from that neighbor's own history. The full design is

$$X = [V, C, Y_1, \ldots, Y_K, N_1, \ldots, N_K]. \tag{2}$$

The paired $Y_i$ and $N_i$ terms expose neighbor residual information. After normalizing each window and channel by its query-context standard deviation, one no-intercept ridge vector is shared across all forecast horizons:

$$\hat{\beta}_\alpha = \arg\min_\beta \|(Y - V) - X\beta\|_2^2 + \alpha\|\beta\|_2^2, \tag{3}$$

$$V' = V + X\hat{\beta}_\alpha. \tag{4}$$

The primary setting is $K = 10$, $\alpha = 10^{-2}$. Validation searches $K \in \{1, 5, 10, 15\}$ and $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$. Delta, convex, per-horizon, and multivariate Adaptime variants are outside the current protocol.

Ridge fitting uses complete adaptation-training rows only; it never applies an element-wise NaN mean to the normal equations. During validation and test, an ineligible query receives the vanilla output for both $C$ and $V'$. This gate uses only query and datastore information available at forecast time. Missing future labels affect metric support only and cannot select the prediction path.

## 3 Evaluation

The official test compares vanilla $V$, retrieval-context $C$ at the selected $K$, and frozen Adaptime $V'$. It retains per-window and per-user MSE, MAE, normalized MSE, and normalized MAE, then reports equal-window and equal-user means and population standard deviations. The TIME-wide view selects only completed task manifests, averages exact repeats before optionally averaging distinct scientific configurations, averages terms within each dataset/frequency, and then gives each dataset/frequency equal weight. Conclusions require completed and inspected cluster artifacts.

Eligibility arrays, reason codes, effective training counts, and test RAG coverage are retained with the task artifacts. The resulting estimand is a transparent gated hybrid: Adaptime on eligible windows and vanilla otherwise.